

CS5285: INTRODUCTION TO CYBERSECURITY

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Introduction to Cybersecurity

Subject Code

CS - Computer Science

Course Number

5285

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

The course aims to provide an understanding of cybersecurity, giving an overview of the requirements and means for the protection of information and transactions, which is an essential feature in the design of any system. The course also

examines a range of security considerations and design issues that are incorporated into the design, development and management of secure systems, including systems for eCommerce.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Identify, explain and apply fundamental cybersecurity services and mechanisms.		x	x	x
2	Demonstrate knowledge of the factors that impact upon the security of systems, and explain different organizational security requirements.		x	x	
3	Apply formulated strategies to assess security of systems and apply appropriate mechanisms to meet security requirements.		x	x	
4	Describe relevant regulations, legal and technical issues governing online transactions and data privacy protection for eCommerce.			x	
5	Design and analyze security protocols to protect systems and data.		x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Students will engage with key concepts of information security. Fundamental principles will be illustrated with real world examples.	1, 2, 3, 4, 5	2 hours/week
2	Tutorials	Students will work on solving information security problems to reinforce understanding of lecture material.	1, 2, 3, 4, 5	1 hour/week

3	Problem Sets	Students will individually apply course concepts to evaluate and create secure systems. Some problems could provide the opportunity to discover how current secure systems operate.	1, 2, 3, 4, 5	4 hr/wk for 4 weeks
---	--------------	---	---------------	---------------------

Additional Information for LTAs

Lectures- CILO No.1-5 (indirectly)

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Problem Set 1	1, 2, 3, 4, 5	10	-	Yes
2	Mid-term Test	1, 2, 3, 4, 5	20	-	No
3	Problem Set 2	1, 2, 3, 4, 5	10	-	Yes

Continuous Assessment (%)

40

Examination (%)

60

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Mid-term Test (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to explain and apply cybersecurity principles.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Problem Sets (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Identify and apply information security principles in evaluating and designing secure systems.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Problem Sets (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Demonstrate ability to identify cybersecurity principles in real-world applications.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Exam (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Ability to describe, analyse and apply concepts related to cybersecurity principles and systems

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Mid-term Test (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to explain and apply cybersecurity principles.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Not even reaching marginal levels

Assessment Task

Problem Sets (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Identify and apply cybersecurity principles in evaluating and designing secure systems.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Not even reaching marginal levels

Assessment Task

Problem Sets (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Demonstrate ability to identify cybersecurity principles in real-world applications.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Not even reaching marginal levels

Assessment Task

Exam (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Ability to describe, analyse and apply concepts related to cybersecurity principles and systems

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

A selection of topics from the following: overview of information security; risks and attacks, security policies and mechanisms; access control, cryptographic techniques, public key infrastructures, authentication and digital certificates; security enforcement in systems; information security management and standards; privacy protection techniques and regulations.

Syllabus:

A selection of topics from the following:

1. Overview of cybersecurity for systems

- Attacks against systems, that include malicious software, network attacks (e.g. DDoS), phishing attack, password guessing attack, etc.

- System protection: firewall, access control mechanisms.

- Security policies for systems, information security management and standards.

- Critique and assessment of security measures.

2. Cryptographic techniques

- Symmetric-key cryptography, public key cryptography
 - Public Key Infrastructure, authentication and digital certificates, electronic transaction ordinance.
3. Protocols and schemes
- Authentication and key management protocols and schemes.
 - Fundamental cryptographic protocols for web/network systems: TLS, IPSec, IKE.
 - Security protocol design

Reading List

Additional Readings

Title	
1	Stallings W. Cryptography and Network Security: Principles and Practice. 8th Ed. Prentice Hall (2022)
2	Stinson D. R. Cryptography - Theory and Practice. 3rd Ed. CRC Press (2005)
3	Anderson R. Security Engineering. 3rd Ed. Wiley (2020)
4	Stamp M. Information Security: Principles and Practice. Wiley (2011)